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TEMPORAL PARTITIONING: AN EXPERIMENT WITH TWO SPECIES OF SPINY MICE

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Abstract. We studied temporal partitioning between two spiny mouse species that coexist in hot rocky deserts in the Middle East: nocturnal common spiny mice (*Acomys cahirinus*) and diurnally active golden spiny mice (*A. russatus*). Although *A. russatus* is diurnally active, it retains the physical activity and body temperature rhythms of nocturnal mammals. We studied the two species in four 1000-m^2 enclosures at Ein Gedi, Israel: two experimental enclosures with *A. russatus* kept alone, and two controls with individuals of both species kept together. We monitored activity with Sherman traps and by studying foraging microhabitat use and efficiency using giving-up densities (GUDs) in food trays. The trays contained broken sunflower seeds mixed in local soil and placed in three microhabitats: under boulders, between boulders, and in the open. Trapping revealed that, in the absence of *A. cahirinus*, the usually diurnal *A. russatus* was active both day and night. However, during the day *A. russatus* still foraged in significantly more patches and to significantly lower GUDs than during the night. Both species, but in particular *A. russatus*, preferred to forage in the boulder habitat. Spiny mice foraged in the same number of trays in the under- and between-boulder microhabitats, but to lower GUDs in the under-boulder microhabitat, both during the day (*A. russatus*) and during the night (both species). The nocturnal *A. cahirinus* exploited more patches with greater efficiency than did *A. russatus* either during the day or during the night. This result suggests that foraging trade-offs that give each species a competitive advantage along some portion of the resource axis cannot be a mechanism of nocturnal coexistence between the two species. Perhaps this is why *A. russatus* resorts to diurnal activity in this hot rocky desert and why the otherwise rare mechanism of temporal partitioning occurs for these species.

Key words: Acomys cahirinus; Acomys russatus; *activity patterns; foraging microhabitat use; giving-up densities; interference competition; mechanisms of coexistence; microhabitat partitioning; rocky deserts; spiny mice; temporal partitioning.*

INTRODUCTION

Ecological theory has long considered niche differentiation in heterogeneous environments to be a major mechanism of coexistence among competing species. Of the many studies that have addressed mechanisms of coexistence between competitors, most have focused on partitioning along the habitat or microhabitat axis (e.g., Rosenzweig 1987, Abramsky et al. 1990, Morris 1996), some have dealt with food-type partitioning (Dayan and Simberloff 1994, Jones 1997, Ben-Moshe et al. 2001), and in recent years a growing number have suggested trade-offs in foraging ecology as a mechanism of coexistence (Brown et al. 1988, Brown 1996, Kotler and Brown 1999). The role of the diel time axis in ecological separation has received little attention (Kronfeld-Schor and Dayan 2003).

Theoretically, temporal partitioning is a viable mechanism of coexistence if the shared limiting resources differ between activity times or if they are renewed within the time frame involved in the separation (Schoener 1974). Schoener's theoretical model suggests that fairly severe resource depletion must occur before it is optimal to stop feeding in a patch frequented by a competitor. He argued that temporal partitioning is rare because ''in deciding to omit certain time periods, the consumer is usually trading something—a lowered but positive yield in the time period frequented by competitors—for nothing, no yield at all'' (Schoener 1974:33). Alternatively, it has been argued that closely related species, prime candidates for competition, are evolutionarily constrained to being active during the same part of the diel cycle (Daan 1981, Kronfeld-Schor et al. 2001*a*, *b*, *c*, Kronfeld-Schor and Dayan 2003).

However uncommon, temporal differences in activity patterns between potential competitors occur in nature, and numerous largely descriptive studies have addressed this phenomenon (reviewed by Kronfeld-Schor and Dayan 2003). On the other hand, experimental work designed to test whether differences in activity patterns actually evolve as a mechanism of coexistence is rare (reviewed by Kronfeld-Schor and Dayan 2003). Ziv et al. (1993) and Kotler et al. (1993) conducted an elegant experiment showing the role of activity patterns among nocturnal desert rodents whose shared resources are renewed daily (see also Brown 1989). Nectar, also a renewable resource, is temporally partitioned by hum-

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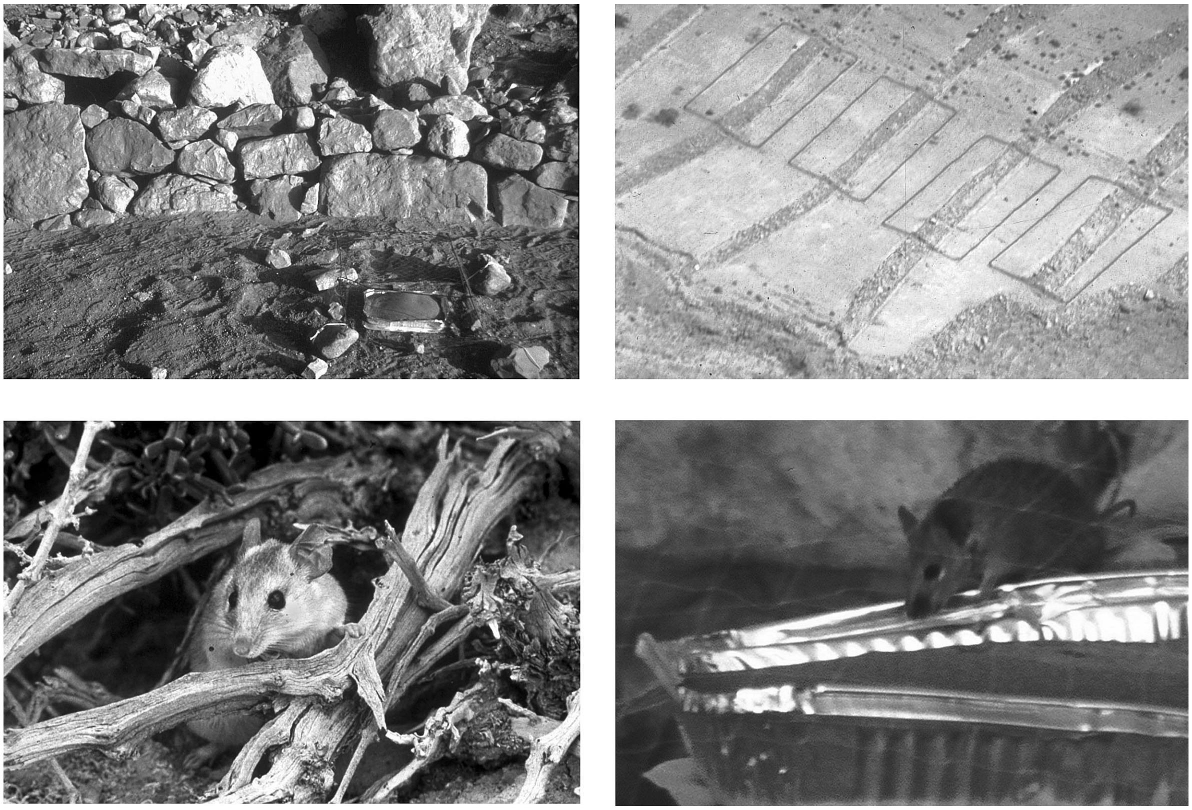

PLATE 1. (Top-left) The three microhabitats: ''under-boulder'' (UB) microhabitat, in which a tray was placed in small crevices that afford shelter for the spiny mice; ''between-boulder'' (BB) microhabitat, in which a tray was placed on the rocky terrace, on top of a surface of smaller stones; and the ''open habitat'' (O) in which a tray was placed in the open ~1 m from the terrace. These three microhabitats constitute a gradient in the degree of shelter and distance from shelter that affects levels of risk from predation while foraging in the patch. Photo credit: Uri Roll. (Top-right) Aerial picture of the four enclosures (20 × 50 m) erected over linear rock terraces (''boulder habitat'') and the adjacent open habitat. Photo credit: Noga Kronfeld-Schor. (Bottom-left) Common spiny mouse (*Acomys cahirinus*). Photo credit: Arieh Landsman. (Bottom-right) Golden spiny mouse (*Acomys russatus*) entering a seed tray through an opening in the fine filament fish netting placed on a heavy wire frame. Photo credit: Uri Roll.

mingbirds (Cotton 1998). In both cases, temporal partitioning is achieved within the same part of the diel cycle (night and day, respectively). No recent experimental work has addressed temporal partitioning between species whose resources differ between night and day.

By carrying out experimental research on temporal partitioning between rocky desert rodents, we wished to gain insight into the role of temporal separation on the diel scale and into the forces underlying its evolution. Shkolnik (1971) showed that, upon removal of the nocturnal common spiny mouse (*Acomys cahirinus*; see Plate 1) from a shared habitat, the diurnally active golden spiny mouse (*A. russatus*; see Plate 1) shifted to nocturnal activity. Thus Shkolnik (1971) concluded that the common spiny mouse competitively excludes the golden spiny mouse and forces it into diurnal activity. This experiment was not replicated at the time, nor repeated at any later date. Activity patterns were monitored only once for 24 hours, and quantitative results were neither published nor analyzed statistically. Although a growing number of physiological and morphological studies find support for viewing the golden spiny mouse as a nocturnal species that has become diurnal (Rubal et al. 1992, Kronfeld-Schor et al. 2001*a*, *b*, *c*), no subsequent research has tested Shkolnik's (1971) hypothesis.

A field study suggested that foraging trade-offs may be a mechanism for coexistence of these mice during the winter months. *A. cahirinus* was a ''cream skimmer,'' a relatively inefficient forager that gave up foraging at relatively high food densities, whereas *A. russatus* was a habitat specialist, perhaps compensating for this restricted niche by foraging very efficiently. However, in summer, the two spiny mouse species overlap to a great extent in their foraging microhabitat use and efficiency (Jones et al. 2001). Both species are omnivorous and subsist on seeds, green vegetation, snails, and arthropods. Insects are a major dietary component, particularly during the summer months (Kronfeld-Schor and Dayan 1999). A recent study of arthropods carried out at our field site revealed that most taxa were either diurnally or nocturnally active; therefore, temporal partitioning may be a viable mechanism of coexistence for their predators (Weinstein 2003).

Jones et al. (2001) hypothesized that temporal partitioning in spiny mice evolved because the more common spatial-habitat partitioning and foraging trade-offs are not viable mechanisms of coexistence during much of the year. This hypothesis, based on results of a field study, can be tested by studying the spatial foraging patterns of both species under experimental conditions, when both are active during the night.

We carried out a controlled and replicated removal experiment in field enclosures, in order to gain insight into the mechanisms of coexistence and into the role of the time axis for this intriguing pair of potentially competing species. Our removal experiment was designed to address the following two issues. First we asked whether removal of common spiny mice affects the activity patterns of golden spiny mice. If such an effect was indeed found, we wanted to gain insight into the evolution of temporal partitioning. Second, we asked whether trade-offs in foraging behaviors (microhabitat use and foraging efficiency) of the two species, when they are active at the same part of the diel cycle, are a mechanism that promotes coexistence.

Materials and Methods

Experimental enclosures

The study was conducted at the Ein Gedi Nature Reserve in the Judean Desert, near the Dead Sea (31°28′ N, 35°23′ E, 300 m below sea level). Four enclosures (20 × 50 m) were erected over linear rock terraces (see Plate 1). They were constructed of 10-mm wire mesh buried 3 cm into the ground and standing 70 cm high. The top 40 cm of both sides of the fence were covered with aluminum flashing to prevent mice from climbing over. However, the natural predators of spiny mice (foxes, snakes, owls, and diurnal raptors) could enter and exit enclosures freely. Because we could not be certain that the food and water naturally available in the enclosures were sufficient to sustain the spiny mice, a limited amount of food (commercial rodent pellets and sunflower seeds) was added every two weeks at the end of each trapping session, and between foraging experiments. Food was not supplemented at the time of the foraging experiments. Two watering troughs in each enclosure provided water throughout the entire period. The troughs were placed in a brick construction with mouse-size entrances, so as not to attract predators and birds to the enclosures.

Populating the enclosures

Spiny mice were captured at the Ein Gedi area, marked individually, and introduced into the enclosures (from which resident individuals had been completly trapped and removed) in the winter of 2000–2001, allowing them several months to acclimate prior to the foraging experiments (June and July 2001). In two control enclosures, we placed eight individuals of each species; in the two experimental enclosures we placed 16 individuals of *Acomys russatus*. Every other week we monitored the time of activity, population size, body mass (using Pesola dynamometer scale to the nearest 2 g), and appearance of young spiny mice, using Sherman live traps that were placed in the enclosures for at least two consecutive days and nights. Individuals of both species can be easily and repeatedly trapped, and exhibit similar trappability (Shargal et al. 2000). We checked traps at sunrise and sunset as follows: we closed them before sunrise and sunset, checked them for trapped rodents, and then reopened them only after sunrise or complete darkness, respectively. By doing so, we could clearly define the time of activity (nocturnal or diurnal). Traps were examined several times throughout the day to prevent dehydration of mice.

Young of both species were removed to other enclosures to regulate population size until ∼6 months prior to the experiment, or released. The sex ratio of the population within the enclosures was also monitored and was managed at ∼1:1 for each species. Foraging experiments began in June 2001 after trapping results indicated a shift in activity patterns of *A. russatus* in the experimental enclosures.

Studying foraging behavior with artificial food patches

We studied the foraging behavior of spiny mice using the giving-up density method (Brown 1988), which assumes that a forager is behaving optimally and that the density of food remaining in the patch when it gives up foraging should correspond to a harvest rate (H) at which the energetic gain from foraging just balances the metabolic cost of foraging (C), the cost of the perceived risk of predation in foraging in that patch (P), and the missed opportunity cost of not foraging elsewhere or indulging in other life maintenance or social activities that have a fitness component (MOC). Thus the quitting harvest rate satisfies $H = C + P + \mathrm{MOC}$. To these components, Kotler and Brown (1988) and Bouskila (1995) add the cost of interference competition (I). GUDs, besides being a measure of quitting harvest rate, are also a measure of foraging efficiency, the extent to which an individual can profitably harvest resources at very low resource abundances (Brown 1988, 1989). During both day and night, standard artificial food patches maintained similar metabolic costs of foraging associated with digging for food and similar missed opportunity costs from not foraging in other artificial food patches. Predation risks during the day and during the night appeared constant across enclosures. Rarely, a viperid snake settled for a day or two in one of the enclosures and had a dramatic effect on foraging activity throughout that single enclosure. The possible cost of interference by *A. cahirinus* over *A. russatus* differed between the control enclosures and the experimental enclosures.

Our artificial food patches consisted of aluminum trays (30 × 20 × 4 cm) containing 2 L of finely sifted local soil and 2 g of crushed and sieved sunflower seed (1–2 mm diameter) mixed thoroughly together (for details of preliminary experiments, see Jones et al. 2001). Frames constructed from heavy wire and fine-filament fish netting kept birds out of the trays during the day (for detailed descriptions, see Jones et al. 2001). Mice reached the trays easily by biting through one strand in the net, but to ensure that any consequent foraging costs were equal across species and treatments, frames were used both day and night and in all microhabitats. Each experiment was preceded by three days and nights of pre-baiting to ensure that the trays had been discovered before the beginning of the experiment.

Nine trays were placed in each enclosure, divided into three stations, three trays in each, one in each microhabitat. The tray stations were placed ∼12 m apart and trays in each station were placed in a triangle ∼3 m apart from one another.

We collected two types of data: (1) number of trays foraged, as a measure of activity level (because spiny mouse numbers were held constant across enclosures); and (2) giving-up densities, which should be independent of the number of mice that have visited the tray (Brown 1988).

Microhabitats studied

Each enclosure contained a rock terrace (''boulder habitat''), on each side of which was natural ''open habitat'' (O) covered with small stones. The boulder habitat comprised jumbled rocks that provide complex microhabitats, with numerous crevices that afford shelter for the spiny mice. In the boulder habitat, we defined the ''under-boulder'' (UB) microhabitat, which afforded overhead shelter, and the ''between-boulder'' (BB) microhabitat, in which a tray was placed on the rocky terrace, on top of a surface of smaller stones, and on at least three sides of which were larger stones between which a mouse could find cover. These two microhabitats, together with the open microhabitat (a tray placed in the open ∼1 m from the terrace), constitute a gradient in the degree of shelter and the distance from shelter that affects levels of risk from predation while foraging in the patch. Note that these microhabitats differ slightly from those of the same definition studied by Jones et al. (2001) and Mandelik et al. (2003) because of enclosure and terrace structure (see *Discussion*).

Experimental protocol

We studied the foraging behavior of *A. russatus* across the diel cycle in the presence and absence of *A. cahirinus*. Moreover, we studied foraging microhabitat use and efficiencies of both species at different parts of the diel cycle and on different parts of the lunar cycle (full moon and new moon).

Foraging experiments were carried out in the summer of 2001 (June and July). Four experiments were conducted: two on full-moon days and nights and two on new-moon days and nights. Each experiment lasted a week: three days of pre-baiting and population monitoring using traps and four days of actual foraging experiments. Trays were examined for footprints and GUDs were measured at sunrise and at sunset, as in Jones et al. (2001). The identity of the foragers in the control enclosures (golden spiny mice or common spiny mice) was inferred from trapping (see *Results*).

Statistical analyses

Trapping during the two months preceding our foraging experiments and during our foraging experiments (overlapping with the pre-baiting stage) consisted of checking the traps at sunset, sunrise, and at four equal intervals during the day. We divided the number of trapped individuals during the day by four for comparison with nocturnal trapping. This is probably a conservative estimate, because in summer there is a midday trough in activity of golden spiny mice in nature (Kronfeld-Schor et al. 2001*b*). We used the chi-square test in order to compare the number of trapping records for each species at each part of the diel cycle.

We compared the body mass of spiny mice in our experimental enclosures with those of free-living spiny mice trapped in the same general region (Shargal et al. 2000), using *t* tests. Analyses were carried out for males only because our experiment was conducted during the reproductive period when female body masses are affected by pregnancy. Individuals were considered adult at 35 g for *A. russatus* and 30 g for *A. cahirinus* (see Shargal et al. 2000). We used the average body mass measured during the last four months of our study.

Influence of the main effects (presence or absence of *A. cahirinus*, lunar phase, species, activity phase, and microhabitat) on the number of foraged trays was analyzed using multi-way frequency analyses and log-linear modeling, after Brown et al. (1988). Following Jones et al. (2001), we fixed all first-order effects (species, moon phase, microhabitat, time of day, and presence or absence of *A. cahirinus*) in the model; only second and higher order effects that included the interaction of the dependent variable (number of trays foraged) with the independent variables were of interest (Sokal and Rohlf 1995). Following Brown et al. (1988), we collapsed data regarding repetitions such as number of enclosure, month of experiment, day of experiment, and location of tray station in the enclosure, in order to avoid empty cells. Additional analyses were done by two-dimensional tests of independence or one-way chi-square tests.

GUD data were arcsine-transformed (arcsine of the square root of proportion; see Brown et al. 1988) to improve their fit to a normal distribution. Because F_{max} tests indicated homogeneity of variances, we used parametric statistics. Experimental design included crossed and nested factors; therefore GUD data were analyzed using partially hierarchal ANOVA, following Kotler et al. (1991) and Brown et al. (1988). Moon phase, presence or absence of *A. cahirinus*, or number of enclosures in part of the statistical analysis were the grouping factors, and microhabitat and time of foraging were the crossed factors. Position of tray station in the enclosure was nested within the *A. cahirinus* presence or absence factor, or the number of enclosures factor, depending on the analysis.

TABLE 1. Influence of microhabitat structure, moon phase, and species or time of activity on the number of trays foraged by *Acomys russatus* and *A. cahirinus* in the control and experimental enclosures; results of final best-fit log-linear model.

Interactions in the best-fit model	Significance of subtracting the term from the best fitted model		
	G	df	*P*
Control enclosures (both species)			
Species† × Moon phase × No. trays foraged	22.932	1	<0.001
Moon phase × No. trays foraged	24.844	2	<0.001
Species† × No. trays foraged	446.643	3	<0.001
Microhabitat × No. trays foraged	438.151	2	<0.001
Species† × Moon phase × Microhabitat	32.513	6	<0.001
Experimental enclosures (only *A. russatus*)			
Time of activity × No. trays foraged	8.903	2	<0.05
Microhabitat × No. trays foraged	233.366	2	<0.001
Microhabitat × Time of activity	12.648	1	<0.001
Nocturnal foraging (*A. russatus* from experimental enclosures vs. *A. cahirinus*)			
Microhabitat × No. trays foraged	197.344	2	<0.001
Species‡ × No. trays foraged	22.961	1	<0.001
Microhabitat × Species	14.165	2	<0.001

† *A. russatus* diurnal foraging vs. *A. cahirinus* nocturnal foraging.
‡ *A. russatus* nocturnal foraging at experimental enclosures vs. *A. cahirinus* nocturnal foraging at control enclosures.

RESULTS

Trapping results revealed that the presence of *Acomys cahirinus* had a significant effect on activity times of *A. russatus*. When kept together, *A. cahirinus* and *A. russatus* were temporally partitioned: *A. cahirinus* was nocturnal while *A. russatus* was diurnal. In four months of trapping, *A. cahirinus* was trapped 102 times during the night and only about five times (adjusted estimate; see statistical analyses in *Materials and Methods*) during the day ($\chi^2 = 73.74$, df = 1, $P < 0.001$). *A. russatus* in the control enclosures was trapped about 54 times during the day and only three times during the night ($\chi^2 = 45.63$, df = 1, $P < 0.001$). In the experimental enclosures, *A. russatus* was trapped more frequently during the night (13 times), but still significantly more frequently during the day (∼63 times) ($\chi^2 = 36.78$, df = 1, $P < 0.001$). A *G* test of independence revealed that the presence of *A. cahirinus* has a significant effect on the time of activity (day or night) of *A. russatus* (with Yates' correction $\chi^2 = 3.27$, df = 1, $P = 0.04$, one-tailed). Additional examination of *A. russatus* night trappings showed that 10 different individuals were trapped during the night (of 13 night trappings), suggesting that nighttime activity was widespread among individuals of *A. russatus* in the experimental enclosures.

T tests revealed that the body mass of male *A. russatus* and *A. cahirinus* in the control (57.8 ± 5.76 g, mean ± 1 SE; $n = 6$; and 43.4 ± 4.04 g, $n = 6$, respectively) and experimental enclosures (48.6 ± 3.75 g, $n = 12$, for *A. russatus* alone) was significantly greater ($P < 0.01$) than those of the free-living populations (free-living *A. russatus* 44.1 ± 0.61 g and free-living *A. cahirinus* 35.2 ± 0.73 g [Shargal et al. 2000]) in the same general area.

In the second new-moon experiment, none of the trays in experimental enclosure number 2 was foraged in at night, owing to the presence of a viper in the enclosure. Therefore, we omitted the number of trays foraged in this experimental enclosure from our analysis. In this specific case we do not have a proper replicate for our nocturnal experiment, and we use stations rather than enclosures as a factor in our analysis.

While enclosed together, *A. cahirinus* foraged in a significantly greater number of trays than did *A. russatus* (species × number of trays foraged in term in log-linear model, Tables 1 and 2, Fig. 1). We detected a nonsignificant general tendency for *A. cahirinus* to use more trays in the BB and O microhabitats; however, there was an interaction of species with moon phase (species × moon phase × number of trays foraged in term in log-linear model, Table 1). During the full-moon phase, there

TABLE 2. Influence of microhabitat structure, moon phase, and species or time of activity on the number of trays foraged by *A. russatus* and *A. cahirinus* in the control and experimental enclosure.

Species and time of activity	Microhabitat			
	BB	O	All	UB
Diurnal *A. russatus* in control enclosures	96	82	7	185
Nocturnal *A. cahirinus* in control enclosures	96	96	26	218
Diurnal *A. russatus* in experimental enclosures	48	48	5	102
Nocturnal *A. russatus* in experimental enclosures	45	40	1	86

Note: Microhabitat abbreviations are: BB, between boulders; O, open habitat; All, all microhabitats; UB, under boulders.

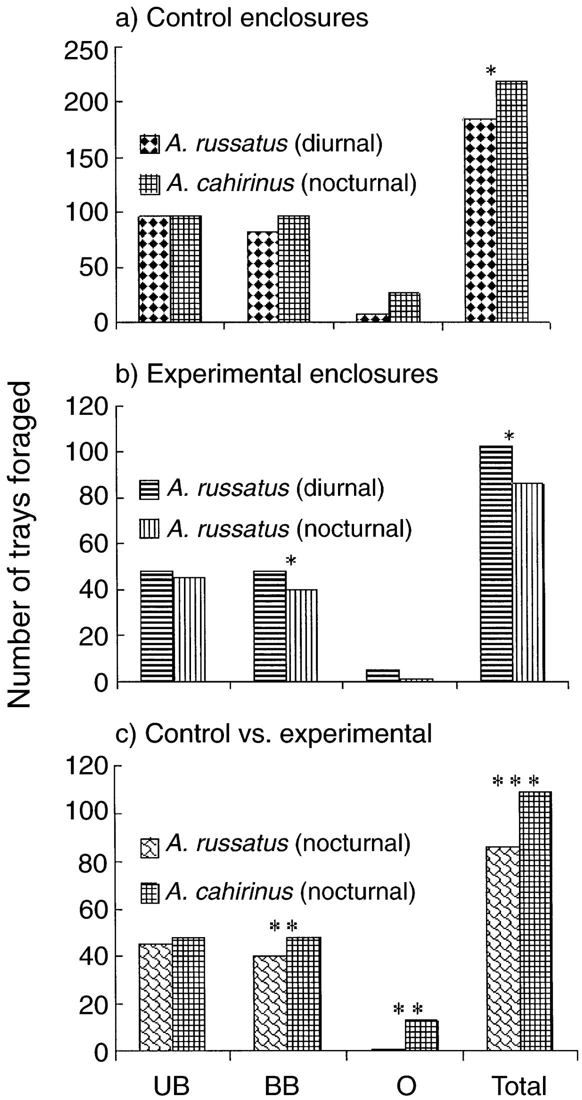

FIG. 1. Number of trays foraged by *A. russatus* and *A. cahirinus* in the control and experimental enclosures by microhabitat (UB, under boulders; BB, between boulders; O, open; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$). In panel (a), the control enclosures, nocturnal foraging is ascribed to *A. cahirinus*, and diurnal foraging to *A. russatus*. In panel (b), the experimental enclosures, both diurnal and nocturnal foraging were carried out by *A. russatus*. Panel (c) compares the *mean* number of trays foraged at night in the control enclosures with the number of trays foraged at night in the experimental enclosures.

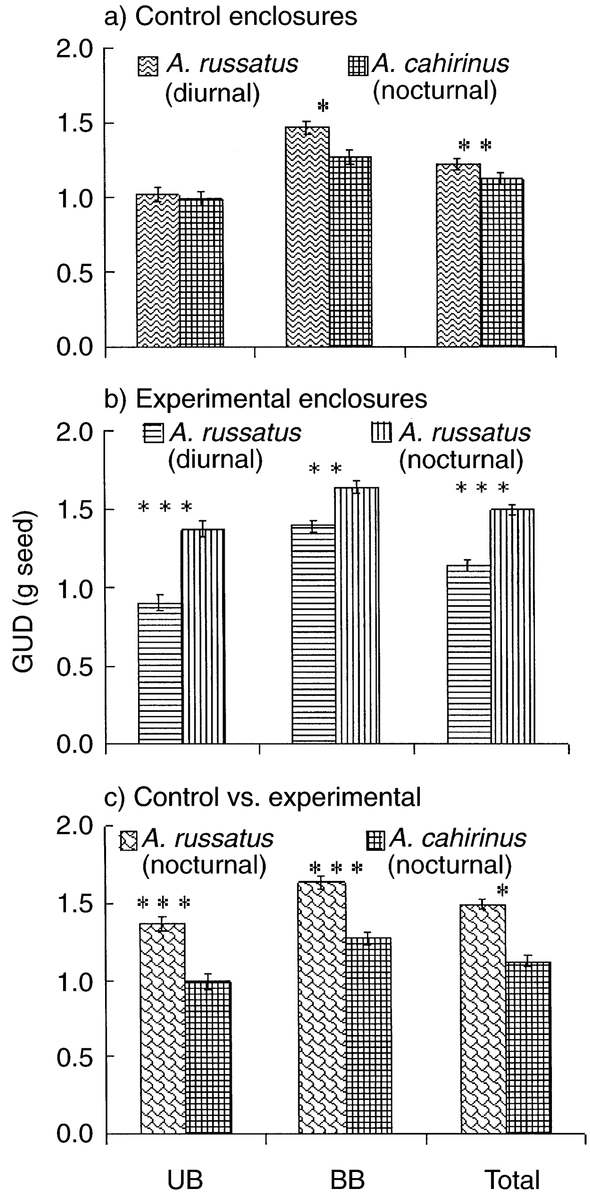

FIG. 2. Mean GUDs of *A. russatus* and *A. cahirinus* in the control and experimental enclosures by microhabitat (UB, under boulders; BB, between boulders; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$). In panel (a), the control enclosures, nocturnal foraging is ascribed to *A. cahirinus*, and diurnal foraging to *A. russatus*. In panel (b), the experimental enclosures, both diurnal and nocturnal foraging were carried out by *A. russatus*. Panel (c) compares GUDs of trays foraged at night in the control enclosures with GUDs of trays foraged at night in the experimental enclosures.

was no significant difference between the number of trays foraged in by each *Acomys* species (*A. russatus* and *A. cahirinus* foraged in 98 and 99 trays, respectively). During new-moon nights, the number of trays foraged in by *A. cahirinus* (199 trays) was significantly greater than that foraged in by *A. russatus* (87 trays) during the following days ($G = 17.46$, df = 1, $P < 0.001$, with Bonferroni correction $k = 2$).

While enclosed together, *A. cahirinus* foraged trays to significantly lower GUDs than did *A. russatus* (Table 3; species term, Table 4; Fig. 2), but there was an interaction with microhabitat structure (species × microhabitat term) so that this general tendency was significant only in the BB microhabitat.

While enclosed without *A. cahirinus*, *A. russatus* foraged a significantly greater number of trays in daytime than during nighttime (time of activity by number of

TABLE 3. The influence of treatment, time of activity, and microhabitat structure on GUD values for *A. russatus* in the experimental enclosures and *A. cahirinus* and *A. russatus* in the control enclosures.

Treatment and species/microhabitat	Mean giving-up density (1 SE)		
	UB	BB	Both
Control enclosures (both species)			
A. russatus (diurnal)	1.02 (0.047)	1.46 (0.045)	1.22 (0.037)
A. cahirinus (nocturnal)	0.99 (0.048)	1.27 (0.045)	1.12 (0.035)
Experimental enclosures (only *A. russatus*)			
Diurnal *A. russatus*	0.90 (0.048)	1.39 (0.036)	1.14 (0.035)
Nocturnal *A. russatus*	1.37 (0.050)	1.64 (0.040)	1.49 (0.034)

trays foraged term in the log-linear model, Tables 1 and 2, Fig. 1), without interaction with microhabitat structure (time of activity × microhabitat × number of trays foraged in term is absent from the model). The foraging efficiency of *A. russatus* was greater during daytime than during nighttime (Table 3; time of activity term, Table 5; Fig. 2) but there was an interaction with type of microhabitat (time of activity × microhabitat term, Table 5; Fig. 2).

At night both species foraged in a greater number of trays in the boulder habitat, without interaction with forager species (absence of species × microhabitat × number of trays foraged term in log-linear model, Tables 1 and 2; and microhabitat term and absence of any interactions, Fig. 1). *A. cahirinus* in the control enclosures foraged in a significantly greater number of trays than did nocturnal *A. russatus* (species × number of trays foraged term in log-linear model, Tables 1 and 2, Fig. 1), a tendency that was significant also in the BB and O microhabitats (proportion of trays foraged; $G = 8.73$, df = 1, $P = 0.0031$, with Bonferroni correction $k = 3$; and $G = 12.04$, df = 1, $P = 0.0005$, with Bonferroni correction $k = 3$ respectively; Fig. 1). Moreover, the foraging efficiency of *A. cahirinus* was significantly greater than that of *A. russatus* (species term, Table 6; Fig. 2) with no interaction with microhabitat structure (no significant species by microhabitat term, Table 6) and was significant even when each microhabitat was analyzed separately (post hoc by Tukey-Kramer method, Fig. 2).

DISCUSSION

In the absence of common spiny mice (*Acomys cahirinus*), golden spiny mice (*A. russatus*) shifted their activity patterns; they foraged both during the day and the night. To our knowledge, this is the first ecological experiment to test temporal partitioning between competitors that can use different resources available during the day and night (Kronfeld-Schor and Dayan 1999).

Because we could not identify spiny mice to the individual level, we cannot tell whether golden spiny mice in the experimental enclosures turned arrhythmic or became active on a shorter rhythmic scale (similar to voles; see Halle 1993), or whether several dominant individuals monopolized the preferred activity time (day or night) while other individuals were excluded to activity during less profitable hours (see Bachmann 1984, Alanara et al. 2001). Nevertheless, although golden spiny mice were nocturnally active in the absence of common spiny mice, much of their activity remained diurnal, and their diurnal foraging remained significantly more efficient than their nocturnal foraging. Our experiment was conducted after four and a half months had elapsed from the time that we populated the experimental enclosures with spiny mice, and during the hot summer months, when golden spiny mice could be expected to shift to cooler nighttime foraging (Shkolnik 1966). Thus the observed temporal activity shift is likely to be robust.

Energetic foraging costs, costs of predation, and missed opportunity costs were held constant for golden

TABLE 4. The influence of species, microhabitat structure, and moon phase on GUD values of foraging *A. russatus* and *A. cahirinus* in the control enclosures (results of partially hierarchical ANOVA on transformed GUD values).

Factor	Type III SS	df	MS	F
Enclosure	2.360×10^{-2}	1	2.360×10^{-2}	0.204
Error 1	0.462	4	0.116	
Microhabitat	4.075	1	4.075	173.088***
Error 2	0.118	5	2.354×10^{-2}	
Moon	1.253	1	1.253	26.402***
Moon × Microhabitat	0.102	1	0.102	2.154
Error 3	0.475	10	4.745×10^{-2}	
Species†	0.426	1	0.426	7.007**
Species† × Microhabitat	0.259	1	0.259	4.258*
Species† × Moon	8.231×10^{-2}	1	8.231×10^{-2}	1.354
Species† × Microhabitat × Moon	0.201	1	0.201	3.307
Error 4	20.487	337	6.079×10^{-2}	

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.
† *A. russatus* diurnal foraging vs. *A. cahirinus* nocturnal foraging.

TABLE 5. The influence of time of activity, microhabitat structure, and moon phase on GUD values for *A. russatus* in the experimental enclosures (results of a partially hierarchical ANOVA on transformed GUD values).

Factor	Type III SS	df	MS	F
Enclosure	0.989	1	0.989	10.628*
Error 1	0.372	4	9.301×10^{-2}	
Microhabitat	3.380	1	3.380	136.274***
Error 2	0.124	5	2.480×10^{-2}	
Moon	0.274	1	0.274	8.934*
Moon × Microhabitat	1.350×10^{-4}	1	1.350×10^{-4}	0.00441
Error 3	0.306	10	3.064×10^{-2}	
Time of activity	3.278	1	3.278	69.552***
Time of activity × Microhabitat	0.265	1	0.265	5.620*
Time of activity × Moon	2.92×10^{-3}	1	2.92×10^{-3}	0.06209
Time of activity × Microhabitat × Moon	9.011×10^{-2}	1	9.011×10^{-2}	1.912
Error 4	13.857	294	4.713×10^{-2}	

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

spiny mice across all four enclosures during the night. The difference between the control and experimental enclosures in the foraging equation (see *Materials and Methods*) lies in the component of interference (*I*; see Kotler and Brown 1988 and Bouskila 1995); it appears that interference (by direct aggression or through resource depletion) by common spiny mice renders nocturnal activity costly to the point of its total absence in golden spiny mice.

Some researchers have emphasized the role of interspecific interference (e.g., Kenagy 1973, Carothers and Jaksic 1984), whereas others have viewed exploitative competition (e.g., Kunz 1973) as more important to temporal partitioning of activity. It is likely that temporal partitioning between the two spiny mouse species in this study has evolved in response to exploitative competition (Kronfeld-Schor and Dayan 1999), but it is difficult to identify the proximate cue limiting golden spiny mice from nocturnal activity. Shkolnik (1971) suggested that the golden spiny mouse is competitively displaced by ''its somewhat more vigorous kindred,'' but under laboratory conditions, golden spiny mice display overt aggression toward common spiny mice (N. Pinter, T. Dayan, N. Kronfeld-Schor, and D. Eilam, *unpublished manuscript*). It previously has been suggested that olfactory cues affect the activity patterns of golden spiny mice (Haim and Rozenfeld 1993, Friedman et al. 1997). Water was available ad lib to spiny mice in all enclosures and food was supplemented between experiments and was plentiful to the extent that individuals of both species were heavier than free-living spiny mice. Seeds in the trays provided additional nutrition during the experiment, so we cannot say whether the shift is resource mediated; additional experiments with varying resource levels are required.

The activity and temperature rhythms of golden spiny mice, their capacity for nonshivering thermogenesis, and their retinal structure are evolutionarily constrained to reflect a nocturnal way of life (Kronfeld-Schor et al. 2001*a*, *b*). Kronfeld-Schor et al. (2001*b*) suggested that temporal partitioning between closely related species, which are usually prime candidates for competition, is rare because of the evolutionary constraints that limit plasticity in the use of different parts of the diel niche axis (Daan 1981, Kronfeld-Schor and Dayan 2003). Why, in the face of these evolutionary constraints, do golden spiny mice shift into diurnal activity?

Under the trade-off theory, coexistence in a habitat and microhabitat mosaic is possible if there are foraging trade-offs between species, either behavioral or

TABLE 6. The influence of species, microhabitat, and moon phase on GUD values foraged *at night* by *A. russatus* in the experimental enclosures and *A. cahirinus* in the control enclosures (results of partially hierarchical ANOVA on transformed GUD values).

Factor	Type III SS	df	MS	F
Species†	3.536	1	3.536	34.433*
Error 1	0.205	2	0.103	
Station	5.169×10^{-2}	2	5.169×10^{-2}	0.346
Error 2	0.448489	6	7.475×10^{-2}	
Microhabitat	1.796	1	1.796	121.869***
Microhabitat × Species†	1.790×10^{-4}	1	1.790×10^{-4}	0.0121
Error 3	0.147333	10	1.473×10^{-2}	
Moon	0.845	1	0.845	15.425***
Moon × Species†	0.116	1	0.116	2.120
Moon × Microhabitat	4.576×10^{-2}	1	4.576×10^{-2}	0.835
Moon × Species† × Microhabitat	9.72×10^{-3}	1	9.72×10^{-3}	0.177
Error 4	16.221	296	0.054800	

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.
† Nocturnal foraging; *A. russatus* at experimental enclosures vs. *A. cahirinus* at control.

evolutionary (morphological or physiological), in the efficiency with which resources can be converted to new biomass (Brown et al. 1994, Vincent et al. 1996), or in relative ability to avoid predation (e.g., Brown et al. 1988, Lima and Dill 1990), or in both.

In the control enclosures, diurnal foraging of golden spiny mice encompassed fewer microhabitats and was less efficient than that of nocturnal common spiny mice. These results support those of Jones et al. (2001), who found that in summer, the two species overlapped to a great extent in their foraging patterns. The slight difference between our results and those of Jones et al. (2001) may reflect subtle differences in microhabitat structure between the enclosures and the field research sites. Be that as it may, both studies show that foraging trade-offs cannot serve as a mechanism of coexistence during the summer months.

Temporal partitioning has evolved from an ancestral state in which both species were nocturnal. In our enclosure experiments, common spiny mice traveled to more food patches than did nocturnally active golden spiny mice. They foraged in more microhabitats and did so more efficiently (to lower giving-up densities) than did golden spiny mice. This implies that, when both species forage during the night, there can be no trade-offs in foraging microhabitat use and efficiencies to enable coexistence between these two species. Thus golden spiny mice may have been driven into diurnal activity in lieu of an alternative mechanism of coexistence (see Jones et al. 2001).

It is interesting to note that, although at least in some aspects of their physiology and morphology, golden spiny mice are nocturnal mammals, when relieved of competition with common spiny mice, they remain active during the day and only extend their activity into the night. It is possible that golden spiny mice have adapted by now to diurnal activity to the extent that they are able to exploit all activity times to their advantage. Dark skin pigmentation, ascorbic acid in their eyes (Koskela et al. 1989), and an efficient water and energy economy (Shkolnik 1966, Degen et al. 1986, Kronfeld-Schor et al. 2001*d*) suggest that this may be so. These adaptations may be the ''ghost of competition past'' (sensu Connell 1980, Rosenzweig 1981), the evolutionary consequence of interspecific competition between these spiny mouse species.

Diurnal and nocturnal activity also expose spiny mice to different predation risks (Jones et al. 2001, Mandelik et al. 2003) and different physiological foraging costs (Kronfeld-Schor et al. 2001*d*), which may affect the choice of activity times (e.g., Wauters and Dhondt 1987, Feener 1988, Lourens and Nel 1990, Flecker 1992, Fenn and MacDonald 1995; reviewed by Kronfeld-Schor and Dayan 2003), and may confer some advantage to diurnal activity in this system. However, analyzing the sum total of evolutionary forces selecting for golden spiny mouse activity patterns is beyond the scope of our study.

In sum, common spiny mice restrict the foraging activity times of golden spiny mice, resulting in temporal partitioning between these competitors. The proximate cause for this shift remains to be studied. Activity patterns of mammals are physiologically and morphologically constrained (Daan 1981, Kronfeld-Schor et al. 2001*b*, Kronfeld-Schor and Dayan 2003). However, the strength of competition between these two desert rodents, coupled with the fact that alternative mechanisms of coexistence cannot operate, suffices to exclude golden spiny mice from nocturnal activity; consequently, temporal partitioning has evolved as a mechanism of coexistence in this rocky desert community.

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